

## A biodemographic analysis using Human Mortality Database data and the Lee–Carter model

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### Abstract

This project investigates mortality trends and future longevity in Luxembourg. The analysis uses age-specific mortality rates and exposures from the Human Mortality Database (HMD) for the period 1960–2024. The research combines period life-table construction, historical life-expectancy estimation, a comparison of female and male longevity, socioeconomic contextualization using GDP per capita from Our World in Data (OWID), and a Lee–Carter mortality forecast for 2025–2054.

The HMD object contains three mortality series—female, male, and total—for ages 0–110 and years 1960–2024. A 2020 period life table based on the total population gives life expectancy at birth of **82.048 years**. The GDP data for Luxembourg contain 36 observations covering 1990–2025, with no missing GDP values in the imported file. The Lee–Carter forecast produces an increasing period life expectancy at birth, from **83.4905 years in 2025** to **88.2810 years in 2054**. A holdout validation, estimating the model with 1990–2014 data and forecasting 2015–2024, produces a mean absolute error of **0.6436 years**.

The results suggest continued longevity improvement under the historical-trend assumptions of the Lee–Carter model. However, the forecast is conditional rather than certain. After excluding 19 years with non-finite sex-specific life-expectancy values, the valid female–male gap reaches a maximum of 7.85 years in 1977 and a minimum of 3.44 years in 2023. The gap is 6.12 years in 1960, 4.57 years in 2020, and 4.55 years in 2024. The main limitations are the replacement of zero mortality rates required for logarithmic modeling, the relatively short 1990–2024 fitting period, the absence of prediction intervals in the reported output, and instability in some early single-sex life-table calculations.

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### 1. Introduction

Mortality analysis is a central part of biodemography because it describes how survival changes over age, calendar time, and sex. It also provides the demographic basis for studying population ageing, longevity, pension sustainability, healthcare demand, and insurance risk. Luxembourg is a particularly interesting case because it is a wealthy European country with high life expectancy and a relatively small population. A small population can produce low or unstable death counts in individual age–year cells, but the country’s high-quality demographic registration makes it suitable for HMD analysis.

This project has two connected components. The first is descriptive: it constructs life tables and examines changes in period life expectancy. The second is predictive: it applies the Lee–Carter mortality model to extrapolate recent

mortality trends into the future.

The analysis is not intended to prove that one socioeconomic variable causes mortality decline. GDP per capita is used as contextual information that helps situate Luxembourg's mortality development within a broader social and economic environment.

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## 2. Purpose, objectives, and research question

### 2.1 Purpose

The purpose of this project is to analyze mortality trends in Luxembourg and to forecast future period life expectancy using a standard stochastic mortality model.

### 2.2 Research question

**How have mortality and period life expectancy changed in Luxembourg between 1960 and 2024, and what future path for life expectancy at birth is implied by a Lee–Carter forecast for 2025–2054?**

### 2.3 Objectives

The project has five objectives:

1. To import and inspect official age-specific mortality and exposure data for Luxembourg.
2. To construct a period life table and explain the meaning of its demographic columns.
3. To calculate the historical development of life expectancy at birth and compare females with males.
4. To contextualize mortality trends using GDP per capita from OWID.
5. To estimate a Lee–Carter model, forecast mortality for 30 years, and evaluate the model using a historical holdout test.

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## 3. Relevance of the project

### 3.1 Pension planning

Life expectancy determines how long pension benefits may need to be paid. If survival improves faster than expected, pension systems face longer payment durations and may require changes in contribution rates, retirement ages, or reserves. A mortality forecast therefore provides a demographic scenario for pension planning.

### 3.2 Healthcare and long-term care

Mortality decline, especially at older ages, is associated with a larger number of people surviving to advanced ages. This can increase demand for medical services, chronic-disease management, rehabilitation, and long-term care. The analysis does not measure morbidity or disability, so higher life expectancy should not automatically be interpreted as longer healthy life expectancy.

### 3.3 Insurance and annuities

Life insurance companies and annuity providers depend on mortality assumptions. A person receiving an annuity for

longer than expected increases the insurer’s liability. Forecasts of age-specific mortality and life expectancy are therefore important for pricing, reserving, and risk management.

3.4 Academic relevance

The analysis applies several core biodemographic concepts covered in the course: period and cohort interpretation, life tables, age-specific mortality, sex differences, mortality modeling, and longevity forecasting.

4. Theoretical framework

4.1 Period and cohort perspectives

A **period** measure uses the mortality rates observed in a particular calendar year or set of calendar years. A period life table describes the hypothetical survival of a synthetic cohort exposed to the mortality schedule of that period.

A **cohort** measure follows people born in the same year through their actual life course. Because the latest cohorts have not yet completed their lives, cohort life expectancy requires long historical follow-up and is not the focus of this project.

The R code uses type = "period", so all reported life-expectancy results are period measures.

Period life expectancy at birth is the expected number of years a hypothetical newborn would live if the current age-specific mortality rates remained unchanged throughout that person's life. It is not the observed average lifespan of a real birth cohort.

4.2 Life-table quantities

A life table begins with age-specific mortality rates and transforms them into probabilities of death and survival quantities.

| Symbol | R output column | Meaning                                               |
|--------|-----------------|-------------------------------------------------------|
| (x)    | x               | Exact age or beginning of the age interval            |
| (m_x)  | mx              | Central death rate at age (x)                         |
| (q_x)  | qx              | Probability of dying between exact ages (x) and (x+1) |
| (l_x)  | lx              | Number surviving to exact age (x)                     |

| Symbol            | R output column | Meaning                                                 |
|-------------------|-----------------|---------------------------------------------------------|
| (d <sub>x</sub> ) | dx              | Number dying between ages (x) and (x+1)                 |
| (L <sub>x</sub> ) | Lx              | Person-years lived between ages (x) and (x+1)           |
| (T <sub>x</sub> ) | Tx              | Total future person-years remaining after exact age (x) |
| (e <sub>x</sub> ) | ex              | Remaining life expectancy at exact age (x)              |

The basic relationships are:  $q_x = \frac{d_x}{l_x}$   $e_x = \frac{T_x}{l_x}$

### 4.3 Demographic transition

The demographic transition framework explains the movement from high mortality and high fertility toward low mortality and low fertility. In the later stages of transition, infant and child mortality are already low, so additional life-expectancy gains increasingly result from reduced adult and old-age mortality. The 2020 life table shows this pattern: mortality is very low during childhood and rises progressively at older ages.

### 4.4 Lee-Carter model

The Lee-Carter model represents the logarithm of the age-specific mortality rate as:  $\log(m_{x,t}) = a_x + b_x k_t + \epsilon_{x,t}$

| Component            | Interpretation                                                     |
|----------------------|--------------------------------------------------------------------|
| (a <sub>x</sub> )    | Average age-specific mortality profile                             |
| (b <sub>x</sub> )    | Sensitivity of mortality at age (x) to the general mortality index |
| (k <sub>t</sub> )    | Time-varying overall mortality level                               |
| ( $\epsilon_{x,t}$ ) | Residual error not explained by the model                          |

The age profile (a<sub>x</sub>) describes how mortality varies across age. The time index (k<sub>t</sub>) summarizes changes in

the overall level of mortality. The forecast extrapolates the time index and reconstructs future age-specific mortality rates. Future mortality rates are then converted into future period life tables and life expectancy.

The Lee-Carter model is useful because it is relatively simple, interpretable, and widely used. Its limitation is that it assumes historical relationships remain sufficiently stable in the future. It cannot automatically anticipate a new epidemic, a major medical innovation, a sudden migration change, or a structural change in health behavior.

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## 5. Data and sources

### 5.1 Human Mortality Database

The primary data source is the Human Mortality Database. The HMD availability table lists Luxembourg with country code **LUX** and period life tables for **1960–2024**, with no listed gap in the period series [1](#). The HMD provides mortality rates, exposures, deaths, populations, and life-table quantities for populations with high-quality demographic registration [2](#).

The R code imports:

```
luxembourg <- read.demogdata(  
  "Mx_1x1.txt",  
  "Exposures_1x1.txt",  
  type = "mortality",  
  label = "Luxembourg",  
  skip = 2  
)
```

The printed R object confirms:

| Data feature                                     | Verified value         |
|--------------------------------------------------|------------------------|
| Country                                          | Luxembourg             |
| HMD code                                         | LUX                    |
| Series                                           | Female, male,<br>total |
| Years                                            | 1960–2024              |
| Ages                                             | 0–110                  |
| Main series used for life tables and forecasting | Total                  |

The total series is selected explicitly with:

```
series_used <- "total"
```

This is preferable to selecting `names(luxembourg$rate)[3]`, because positional selection could fail if the order of series changes.

## 5.2 Our World in Data

The socioeconomic information comes from the OWID file `access-to-clean-fuels-for-cooking-vs-gdp-per-capita.csv`. The project uses the variable `ny_gdp_pcap_pp_kd`, which is *GDP per capita in purchasing-power-parity terms, measured in constant 2021 international dollars* (OWID, 2026).

The imported Luxembourg subset has the following verified characteristics:

| OWID characteristic       | Result                         |
|---------------------------|--------------------------------|
| Number of Luxembourg rows | 36                             |
| First year                | 1990                           |
| Last year                 | 2025                           |
| Missing GDP values        | 0                              |
| GDP indicator             | <code>ny_gdp_pcap_pp_kd</code> |

The GDP series begins in 1990, so it cannot be compared with the full HMD period beginning in 1960. The GDP comparison is therefore interpreted over the overlapping period, approximately 1990–2024 for the mortality analysis.

The GDP graph is descriptive. Both GDP per capita and life expectancy may rise over time, but two upward-trending series do not demonstrate that GDP caused the change in life expectancy. A causal analysis would require a different research design, additional variables, and statistical modeling.

## 5.3 Source documentation required for the final version

| Source | Data                          | Years     |
|--------|-------------------------------|-----------|
| HMD    | Mortality rates,<br>exposures | 1960-2024 |

| Source | Data           | Years     |
|--------|----------------|-----------|
| OWID   | GDP per capita | 1990-2025 |

The OWID file was downloaded on 14 August 2026.

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## **6. Methodology and R code explanation**

### **6.1 Working directory and package**

The script begins by setting the working directory and loading the demography package. The package should be installed once before analysis, rather than installed every time the script runs.

```
library(demography)
```

### **6.2 Importing mortality and exposure data**

`read.demogdata()` reads mortality rates and exposures by age and year. Mortality rates alone are insufficient for many demographic calculations because the exposure population is needed to understand the population at risk. The `skip = 2` argument skips the two header lines in the HMD text file.

The script checks the available series and prints the data object. This is important because it confirms that the imported data contain the expected sex-specific and total series and the expected age and year ranges.

### **6.3 Constructing the complete period life table**

The code uses:

```
lux.lt <- lifetable(

  luxembourg,
  series = series_used,
  years  = luxembourg$year,
  ages   = luxembourg$age,
  max.age = 100,
  type   = "period"

)
```

The purpose is to calculate life-table quantities for every available year. The table is restricted to age 100 because very old ages in a small population may contain few observations and unstable rates.

#### 6.4 Constructing the 2020 life table

The 2020 table is a detailed example used to demonstrate the calculation and interpretation of life-table columns:

```
lux_2020.lt <- lifetable(  
  luxembourg,  
  series = series_used,  
  years = 2020,  
  ages = luxembourg$age,  
  max.age = 100,  
  type = "period"  
)
```

The selected output rows below show the beginning, middle, and end of the life table.

| Year | Age | mx     | qx     | lx     | dx     | Lx     | Tx      | ex      |
|------|-----|--------|--------|--------|--------|--------|---------|---------|
| 2020 | 0   | 0.0045 | 0.0045 | 1.0000 | 0.0045 | 0.9958 | 82.0480 | 82.0480 |
| 2020 | 1   | 0.0000 | 0.0000 | 0.9955 | 0.0000 | 0.9955 | 81.0522 | 81.4159 |
| 2020 | 20  | 0.0000 | 0.0000 | 0.9930 | 0.0000 | 0.9930 | 62.1482 | 62.5853 |
| 2020 | 40  | 0.0009 | 0.0009 | 0.9860 | 0.0009 | 0.9856 | 42.3409 | 42.9405 |
| 2020 | 50  | 0.0029 | 0.0028 | 0.9771 | 0.0028 | 0.9757 | 33.5001 | 34.2852 |
| 2020 | 60  | 0.0055 | 0.0055 | 0.9417 | 0.0052 | 0.9391 | 22.9091 | 24.3279 |
| 2020 | 65  | 0.0112 | 0.0111 | 0.9098 | 0.0101 | 0.9047 | 18.2705 | 20.0821 |
| 2020 | 70  | 0.0150 | 0.0149 | 0.8575 | 0.0128 | 0.8512 | 13.8443 | 16.1442 |
| 2020 | 80  | 0.0414 | 0.0406 | 0.6648 | 0.0270 | 0.6513 | 6.0965  | 9.1704  |
| 2020 | 90  | 0.1388 | 0.1297 | 0.3253 | 0.0422 | 0.3042 | 1.4976  | 4.6043  |

| Year | Age | mx     | qx     | lx     | dx     | Lx     | Tx     | ex     |
|------|-----|--------|--------|--------|--------|--------|--------|--------|
| 2020 | 100 | 0.5330 | 1.0000 | 0.0191 | 0.0191 | 0.0358 | 0.0358 | 1.8762 |

The key result is:

Luxembourg’s total period life expectancy at birth in 2020 was 82.048 years.

The table shows that mortality is very low in childhood and early adulthood. Survival remains high up to middle age, but mortality increases considerably after age 60 and especially after age 80. This is the expected pattern in a low-mortality population.

6.5 Historical life expectancy

The code calculates the total period life-expectancy series:

```
lux_e0 <- life.expectancy(  
  
  luxembourg,  
  series = series_used,  
  years = luxembourg$year,  
  type = "period"  
)  
  
plot(lux_e0, main = "Life Expectancy at Birth. Luxembourg, 1960-2024",  
  xlab = "Year", ylab = "e0 (years)")
```

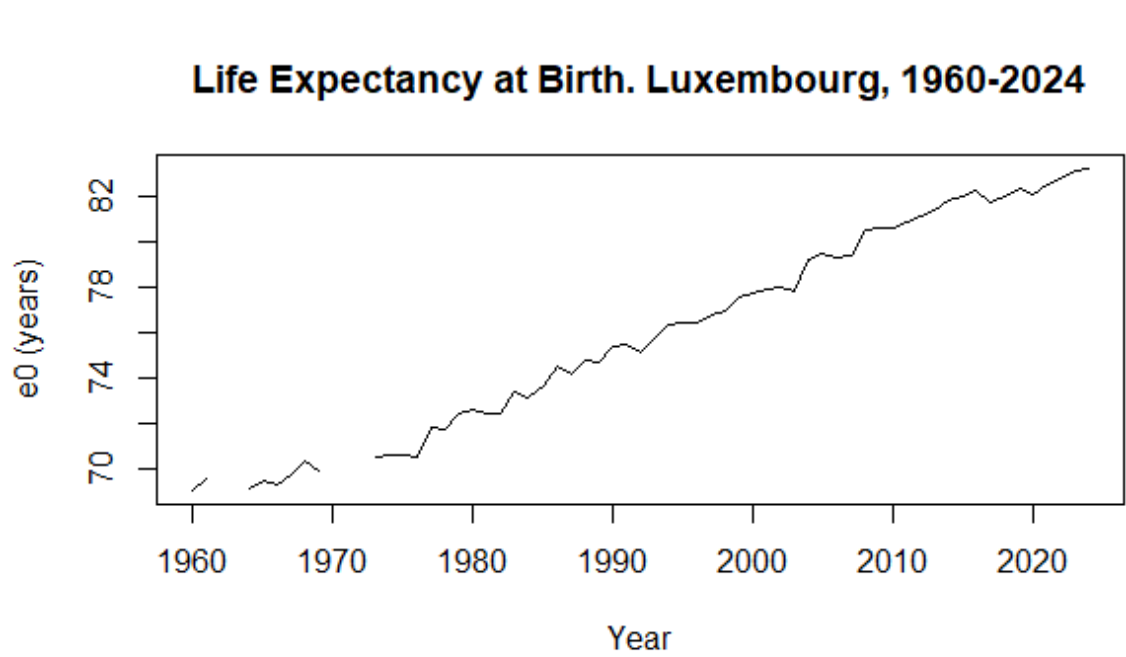


Figure 1: Life expectancy

xpectancy at birth in Luxembourg, 1960-2024.  
Source: HMD

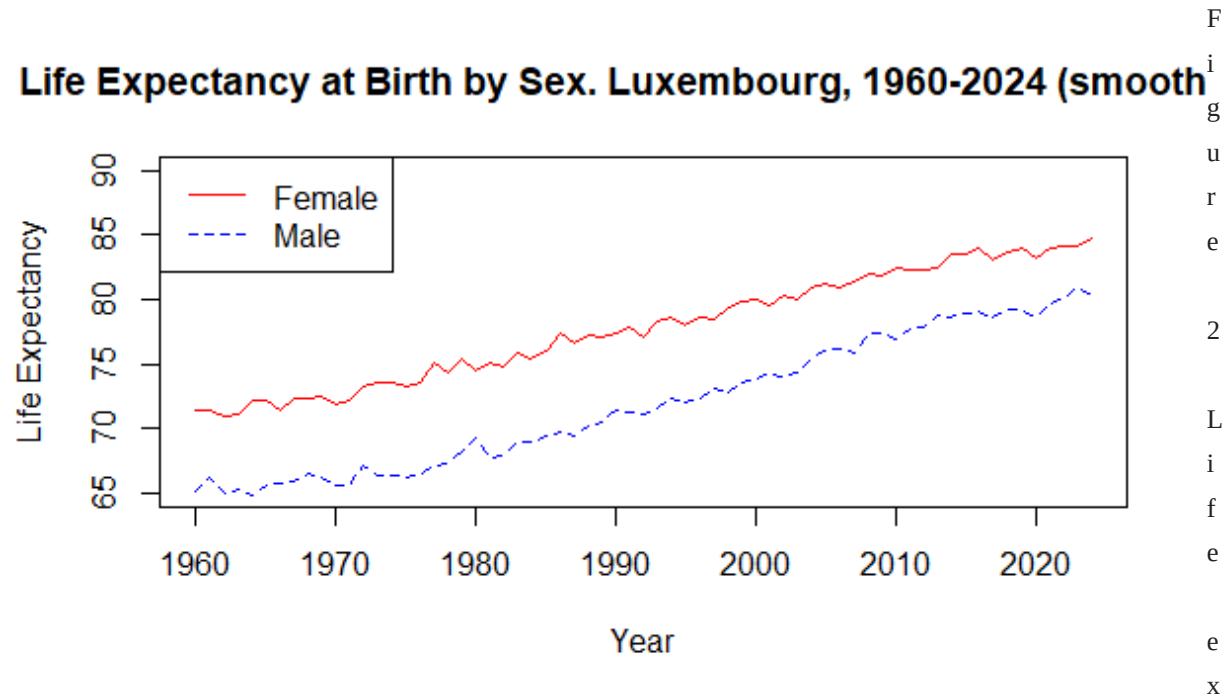
The corresponding graph should show the long-run trend from 1960 to 2024. The graph is important because it converts the complex age-specific mortality surface into a single interpretable longevity indicator.

6.6 Female and male life expectancy

The code calculates separate female and male series:

```
e0_female <- life.expectancy(luxembourg, series = "female",  
  
                             years = luxembourg$year, type = "period")  
  
e0_male <- life.expectancy(luxembourg, series = "male",  
  
                           years = luxembourg$year, type = "period")  
  
plot(e0_female, col = "red", ylim = c(65, 90), xlab = "Year",  
     ylab = "Life Expectancy",  
     main = "Life Expectancy at Birth by Sex. Luxembourg, 1960-2024")  
lines(e0_male, col = "blue", lty = 2)  
legend("topleft", c("Female", "Male"), col = c("red", "blue"), lty = 1:2)
```

Life Expectancy at Birth by Sex. Luxembourg, 1960-2024 (smooth



Source: HMD

The female–male gap is calculated as:

$$[ \text{Gap}_t = e_{\{0,t\}^{\{\text{female}\}}} - e_{\{0,t\}^{\{\text{male}\}}} . ]$$

The first calculation of the sex gap produced non-finite values in 19 years. These values are not interpretable as real life expectancy. They arise from unstable or undefined sex-specific life-table calculations in a small population, where some age–year cells have very sparse mortality or exposure information. The analysis therefore keeps only finite female, male, and gap values before calculating summaries.

```
sex_gap_finite <- sex_gap[
  is.finite(sex_gap$e0_female) &
  is.finite(sex_gap$e0_male) &
  is.finite(sex_gap$gap_F_minus_M),
]
```

```
> cat("\nMaximum female-male gap (valid years only):\n"); print(max_gap_row)
```

Maximum female-male gap (valid years only):

|    | year | e0_female | e0_male  | gap_F_minus_M |
|----|------|-----------|----------|---------------|
| 18 | 1977 | 75.77746  | 67.92759 | 7.849872      |

```
> cat("\nMinimum female-male gap (valid years only):\n"); print(min_gap_row)
```

Minimum female-male gap (valid years only):

|    | year | e0_female | e0_male  | gap_F_minus_M |
|----|------|-----------|----------|---------------|
| 64 | 2023 | 84.86335  | 81.42194 | 3.441415      |

The cleaned results show a maximum valid female–male gap of 7.85 years in 1977 and a minimum valid gap of 3.44 years in 2023. The gap declined from 6.12 years in 1960 to 4.55 years in 2024. Nineteen years were excluded from the sex-gap summaries, and this exclusion is reported as a data-quality limitation rather than concealed.

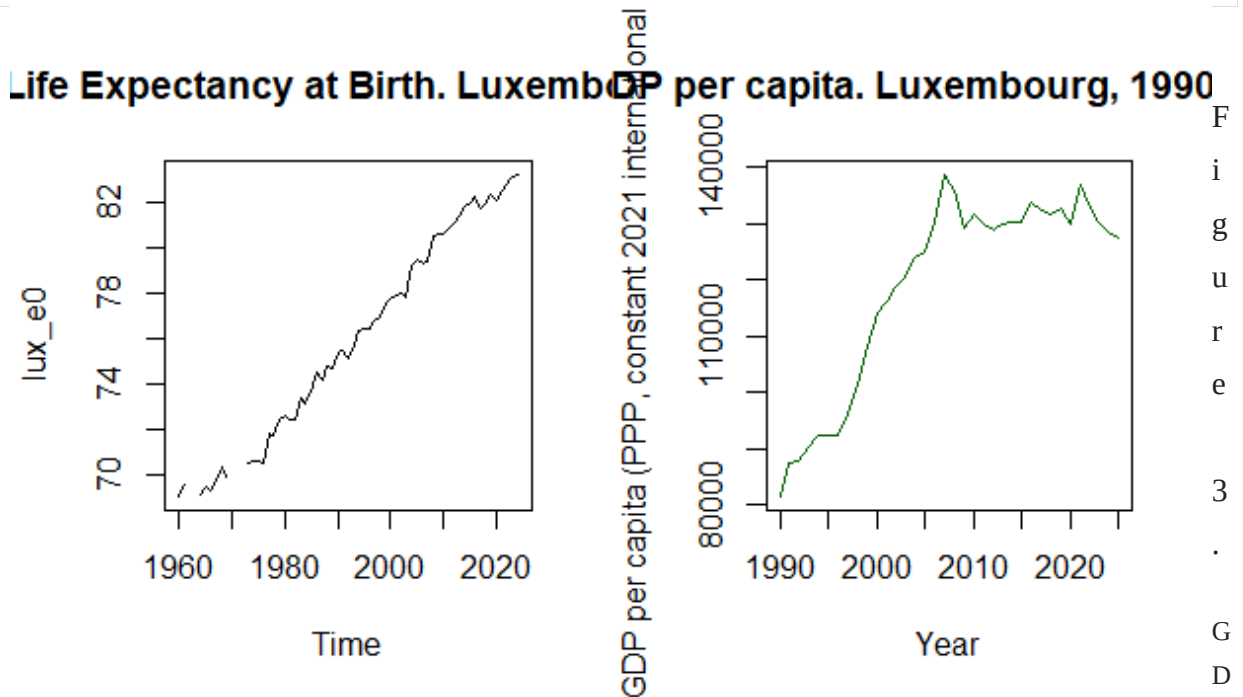
## 6.7 GDP comparison

The GDP data are imported and filtered to Luxembourg. The output shows 36 observations from 1990 to 2025 and no missing GDP values. The side-by-side graph places total life expectancy and GDP per capita next to each other.

```
owid <- read.csv("access-to-clean-fuels-for-cooking-vs-gdp-per-capita.csv")
lux_owid <- subset(owid, entity == "Luxembourg")
```

```
cat("\nLuxembourg OWID rows: ", nrow(lux_owid), "\n") cat("GDP data available from year: ",
min(lux_owid$year), " to ", max(lux_owid$year), "\n") cat("Missing GDP values: ",
sum(is.na(lux_owid$ny_gdp_pcap_pp_kd)), " of ", nrow(lux_owid), " rows\n")
```

```
par(mfrow = c(1, 2)) plot(lux_owid$year, lux_owid$ny_gdp_pcap_pp_kd, type = "l", col = "darkgreen", xlab =
"Year", ylab = "GDP per capita (PPP, constant 2021 international $)", main = "GDP per capita. Luxembourg, 1990-
2025") par(mfrow = c(1, 1))
```



P per capita and life expectancy at birth, Luxembourg, 1990-2024. Source: OWID & HMD.

This graph is included because it provides the required quantitative socioeconomic information. It helps describe the context in which mortality improvements occurred. However, it is not a regression, and the project does not estimate a causal effect of GDP on life expectancy.

## 6.8 Lee–Carter estimation

The Lee–Carter model is fitted using years 1990–2024 and ages 0–100:

```
lux.LC <- lca(
luxembourg,
series = series_used,
years = 1990:2024,
ages = 0:100,
interpolate = TRUE
)
```

The model is fitted to the recent period rather than the complete 1960–2024 period because mortality conditions, medical technology, and demographic regimes may differ substantially between the 1960s and the recent decades. The cost of this choice is that fewer years are used, which can increase uncertainty.

## Luxembourg: total death rates (2025-2054)

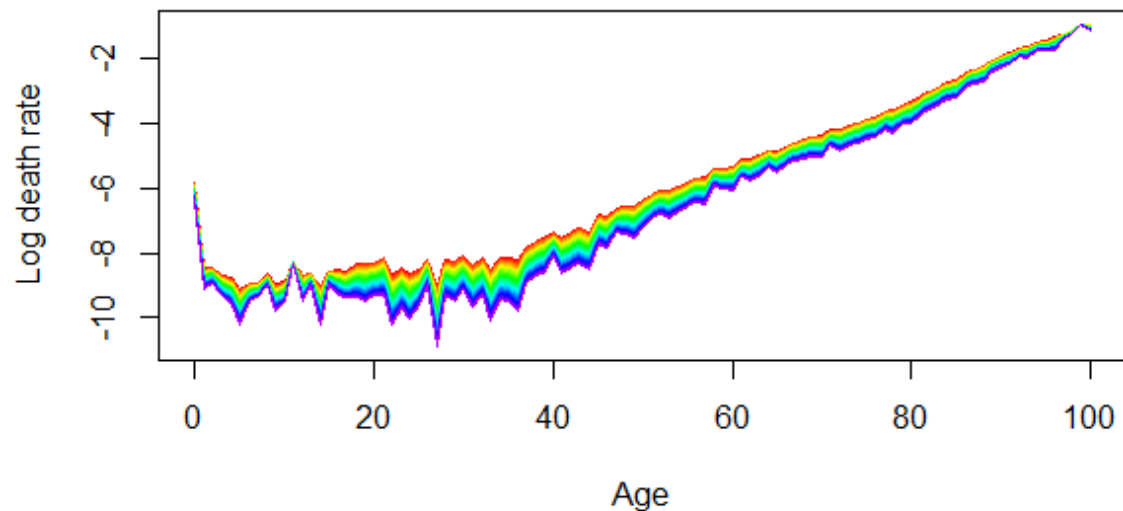


Figure 4 .  
Forecast

**ecast age-specific log mortality rates for Luxembourg, 2025–2054.** The curves show the Lee–Carter projected logarithmic death rates by age for each forecast year. The x-axis represents age and the y-axis represents log mortality rates. The figure illustrates the projected age pattern of mortality over the 30-year forecast horizon.

The model generates the warning:

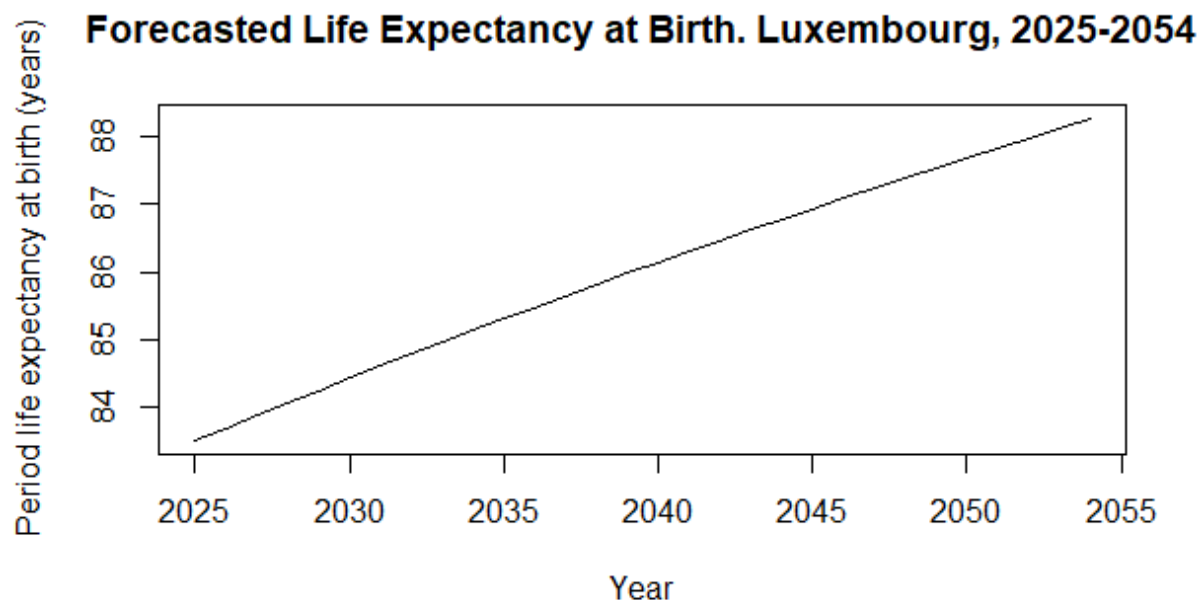
```
Replacing zero values with estimates
```

This warning is important. The Lee–Carter model uses logarithmic mortality rates, but  $(\log(0))$  is undefined. Some age–year cells contain zero observed mortality rates, especially where the number of deaths is small. The `interpolate = TRUE` setting replaces or estimates these values so that the model can be fitted. Therefore, the fitted model is based partly on estimated values rather than entirely on directly observed positive mortality rates.

### 6.9 Forecasting

The model is forecast 30 years ahead:

```
lux.forecast <- forecast(lux.LC, h = 30)
lux_e0_forecast <- life.expectancy(lux.forecast, type = "period")
```



sted period life expectancy at birth for Luxembourg, 2025-2054. Source: Author's calculations based on the Lee-Carter model and HMD data.

The forecast is converted into period life expectancy for each year from 2025 to 2054.

## 7. Results

### 7.1 Main life-table result

The 2020 total period life table gives:

The value  $q_x = 1.0000$  at age 100 does not represent a normal one-year death probability; it reflects the closure convention for the final open age interval (100+), where all remaining survivors are assumed to die within that interval by construction.

| Indicator                                                 | Value                   |
|-----------------------------------------------------------|-------------------------|
| Period life expectancy at birth, ( $e_0$ )                | <b>82.0480</b><br>years |
| Infant mortality rate, ( $m_0$ )                          | 0.0045                  |
| Probability of dying before age 1, ( $q_0$ )              | 0.0045                  |
| Survivors at age 65, ( $l_{\{65\}}$ )                     | 0.9098                  |
| Probability of dying from age 99 to 100, ( $q_{\{99\}}$ ) | 0.3449                  |

| Indicator                               | Value  |
|-----------------------------------------|--------|
| Terminal probability at age 100, (q100) | 1.0000 |

The life table is capped at age 100, which is treated as the final open age interval (100+ years). The value  $q_{100} = 1.0000$  is a mathematical closure convention, not an observed probability of death between ages 100 and 101. This convention ensures that the life table sums correctly and all survivors are eventually accounted for.

The value of 82.0480 years is a period measure. It does not mean that every person born in Luxembourg in 2020 will live exactly 82.048 years. It means that a synthetic cohort exposed throughout life to the 2020 mortality schedule would have that average length of life.

## 7.2 GDP results

| Indicator                   | Result                                               |
|-----------------------------|------------------------------------------------------|
| Luxembourg GDP observations | 36                                                   |
| GDP period                  | 1990–2025                                            |
| Missing GDP observations    | 0                                                    |
| Variable used               | ny_gdp_pcap_pp_kd                                    |
| Unit described in script    | GDP per capita (PPP, constant 2021 international \$) |

The GDP series is suitable for descriptive context. Because the GDP data begin in 1990, the GDP comparison does not cover the earlier 1960–1989 mortality period.

## 7.3 Lee–Carter forecast table

| Year | Forecast period (e_0) |
|------|-----------------------|
| 2025 | 83.49052              |
| 2026 | 83.67957              |
| 2027 | 83.86672              |
| 2028 | 84.05200              |
| 2029 | 84.23545              |

| Year | Forecast period (e_0) |
|------|-----------------------|
| 2030 | 84.41707              |
| 2031 | 84.59689              |
| 2032 | 84.77493              |
| 2033 | 84.95122              |
| 2034 | 85.12577              |
| 2035 | 85.29861              |
| 2036 | 85.46974              |
| 2037 | 85.63921              |
| 2038 | 85.80701              |
| 2039 | 85.97317              |
| 2040 | 86.13772              |
| 2041 | 86.30066              |
| 2042 | 86.46202              |
| 2043 | 86.62181              |
| 2044 | 86.78005              |
| 2045 | 86.93676              |
| 2046 | 87.09196              |
| 2047 | 87.24566              |
| 2048 | 87.39787              |
| 2049 | 87.54863              |

| Year | Forecast period (e_0) |
|------|-----------------------|
| 2050 | 87.69793              |
| 2051 | 87.84580              |
| 2052 | 87.99225              |
| 2053 | 88.13730              |
| 2054 | 88.28096              |

The forecast increases from 83.49052 years in 2025 to 88.28096 years in 2054, an increase of approximately 4.79044 years over the forecast horizon. The average increase is approximately 0.165 years per year, although the annual increments gradually become smaller in the displayed sequence.

This pattern represents continued improvement under the model. It should not be interpreted as a guaranteed demographic outcome because the model does not include prediction intervals in the displayed output and cannot foresee structural breaks.

#### 7.4 Holdout validation results

The validation model is fitted using 1990–2014 data and forecasts 2015–2024. The forecast is compared with the observed HMD-derived period life expectancy.

| Year | Actual (e_0) | Predicted (e_0) | Actual – predicted |
|------|--------------|-----------------|--------------------|
| 2015 | 81.97126     | 82.07713        | -0.10587           |
| 2016 | 82.28345     | 82.30228        | -0.01883           |
| 2017 | 81.76563     | 82.52473        | -0.75910           |
| 2018 | 82.02586     | 82.74453        | -0.71868           |
| 2019 | 82.38174     | 82.96173        | -0.57999           |
| 2020 | 82.04797     | 83.17636        | -1.12839           |
| 2021 | 82.48433     | 83.38846        | -0.90413           |
| 2022 | 82.80261     | 83.59808        | -0.79547           |

| Year | Actual (e_0) | Predicted (e_0) | Actual – predicted |
|------|--------------|-----------------|--------------------|
| 2023 | 83.15367     | 83.80524        | -0.65157           |
| 2024 | 83.23587     | 84.00999        | -0.77412           |

The mean absolute error is:

[  $MAE = 0.6436 \text{ years}$ . ]

The predicted values are higher than the observed values in every validation year, because the error is defined as actual minus predicted and is negative throughout. The largest error occurs in 2020, when the predicted value exceeds the observed value by approximately 1.128 years. The model also overpredicts 2021–2024 by approximately 0.65–0.90 years.

This result is important for interpretation. The Lee–Carter model captures the general upward longevity trend, but it does not reproduce every short-term fluctuation. The validation period includes years affected by unusual mortality conditions, so the forecast should be treated as a baseline trend scenario rather than an exact prediction.

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## **8. Discussion**

### **8.1 Mortality pattern**

The 2020 life table shows the typical profile of a low-mortality population. Mortality is very low during childhood and early adulthood, survival remains high through middle age, and mortality increases rapidly at older ages. This pattern is consistent with the later stage of demographic transition, where the main remaining mortality burden is concentrated in older ages.

### **8.2 Longevity and social planning**

A period life expectancy of approximately 82 years indicates a high-longevity setting. Continued improvement to approximately 88.3 years by 2054 under the Lee–Carter scenario would have important consequences for pension duration, health-service planning, and insurance products. However, the forecast concerns period life expectancy; it is not a direct forecast of the actual final lifespan of a specific cohort.

### **8.3 Socioeconomic context**

The GDP series provides evidence that Luxembourg’s mortality analysis occurs in a high-income context, but the graph alone cannot identify a causal mechanism. Economic development may affect health through income, housing, nutrition, education, healthcare access, and working conditions. Nevertheless, the present project is descriptive and does not include a causal regression.

### **8.4 Sex differences**

After excluding 19 years with non-finite female or male life-expectancy values, the valid sex-gap results are summarized below. The gap is defined as female life expectancy minus male life expectancy.

| Result            | Year | Female (e_0)      | Male (e_0) | Gap               |
|-------------------|------|-------------------|------------|-------------------|
| Maximum valid gap | 1977 | 75.78             | 67.93      | <b>7.85 years</b> |
| Minimum valid gap | 2023 | 84.86             | 81.42      | <b>3.44 years</b> |
| Reference year    |      | Gap               |            |                   |
| 1960              |      | <b>6.12 years</b> |            |                   |
| 1990              |      | <b>5.94 years</b> |            |                   |
| 2020              |      | <b>4.57 years</b> |            |                   |
| 2024              |      | <b>4.55 years</b> |            |                   |

The results indicate a substantial narrowing of the female–male life-expectancy gap over the valid observation period: from 6.12 years in 1960 to 4.55 years in 2024, with the widest observed gap in 1977 and the narrowest valid gap in 2023. The 19 excluded years must remain documented because the exclusion reflects unstable or undefined single-sex life-table calculations in the small Luxembourg population.

### 8.5 Forecast performance

The holdout MAE was 0.6436 years over the 2015–2024 validation period. This means that the forecasts differed from observed life expectancy by an average absolute amount of approximately 0.64 years over this particular ten-year holdout. The model follows the general trend but overestimates life expectancy in each validation year. This systematic pattern suggests that the recent mortality period did not evolve exactly as the model projected from 1990–2014. The result supports presenting the 2025–2054 output as a conditional scenario with uncertainty rather than as a guaranteed forecast.

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## 9. Limitations

### 9.1 Zero mortality rates

The Lee–Carter model works with logarithmic mortality rates. Zero rates cannot be logged, so the `demography` package replaces some zeros with estimates. This is necessary computationally but introduces preprocessing uncertainty, especially at young ages and in a small population.

### 9.2 Small population size

Luxembourg’s small population means that some age–year–sex cells contain few deaths. This can generate zero rates, unstable rates, or unusual period estimates. The age-100 cap is a practical way to avoid modeling the most unstable extreme ages, but it also means the open interval beyond age 100 is simplified.

### 9.3 Fitting-period choice

The Lee–Carter model is fitted only from 1990 to 2024. This provides a more recent mortality regime but uses fewer observations than a 1960–2024 model. The choice should be treated as a modeling assumption and tested in future work using alternative fitting windows.

### 9.4 No prediction intervals shown

The reported forecast is a point forecast. Prediction intervals are not included in the displayed output, so uncertainty around the 2025–2054 forecasts is not quantified. The forecast should therefore be interpreted as a conditional scenario rather than a precise prediction.

### 9.5 Socioeconomic analysis is descriptive

GDP per capita is included as context, but no regression or causal analysis is conducted. The project should not conclude that GDP caused the improvement in life expectancy.

### 9.6 Non-finite sex-specific life-expectancy values

The preliminary sex-gap calculation produced Inf and -Inf values in 19 years, not only in 1961–1962. These values are not meaningful demographic estimates. They arise from unstable or undefined sex-specific life-table calculations in some years, likely related to sparse single-sex mortality or exposure cells at particular ages. The analysis therefore filters to finite female, male, and gap values before calculating the maximum, minimum, selected-year results, or any trend interpretation. The number of excluded years is reported transparently rather than hidden.

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## 10. Conclusions

This project applies appropriate biodemographic methods to real mortality data for Luxembourg. The HMD dataset contains female, male, and total mortality series covering 1960–2024 and ages 0–110. The life-table analysis shows a 2020 total period life expectancy at birth of 82.048 years. Mortality is low at young ages and rises substantially at older ages, as expected in a high-longevity population.

The socioeconomic part of the project adds GDP per capita from OWID for 1990–2025. The data contain 36 Luxembourg observations and no missing GDP values. This information provides useful context but is descriptive and should not be interpreted as evidence of causality.

The Lee–Carter model estimated from 1990–2024 forecasts period life expectancy rising from 83.4905 years in 2025 to 88.2810 years in 2054. The holdout validation has a mean absolute error of 0.6436 years, with systematic overprediction during 2015–2024. This confirms that the model captures the broad direction of longevity but does not perfectly predict short-term changes.

The valid sex-specific results show that the female–male life-expectancy gap narrowed from 6.12 years in 1960 to 4.55 years in 2024. The maximum valid gap was 7.85 years in 1977, while the minimum valid gap was 3.44 years in 2023. Nineteen years were excluded because one or more sex-specific life-expectancy values were non-finite. These exclusions are a limitation caused by sparse single-sex data, but documenting them strengthens the transparency of the analysis.

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Note: GDP per capita is expressed in international-\$ at 2021 prices.

## Appendix. Important R output values

| Output                    | Value         |
|---------------------------|---------------|
| HMD years                 | 1960–2024     |
| HMD ages                  | 0–110         |
| 2020 total period (e_0)   | 82.0480       |
| OWID Luxembourg rows      | 36            |
| OWID years                | 1990–2025     |
| Missing GDP observations  | 0             |
| Lee–Carter fitting years  | 1990–2024     |
| Lee–Carter forecast years | 2025–2054     |
| Forecast (e_0) in 2025    | 83.49052      |
| Forecast (e_0) in 2054    | 88.28096      |
| Forecast increase         | 4.79044 years |
| Holdout period            | 2015–2024     |

| Output                      | Value        |
|-----------------------------|--------------|
| Holdout mean absolute error | 0.6436 years |